SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Vignesh SH/192412046 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Vignesh

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

CO1 Assessment Tool 3

1. Intelligent Agent in a Smart home Automation system

Introduction:

An Intelligent agent is a system that observe its environment through sensor and take suitable action using actuators to acheive specific goals. In smart home it control lighting temperature & security based on user behaviour & environmental conditions

1. Simple Reflex Agent

A simple reflex agent make decisions based only on the current input using condition action rule

- If motion is detected the light turn ON
- If the temperature is high the AC tu
- If an intruder is detected the alarm activate

Advantage:

Fast, Simple, & easy to implement

partial information
al environment 3007

Disadvantage:

Cannot remember past event or lean user preferences

2. Model Based agent

A model based agent maintains as internal state using current & previous observation

- Remembers whether a room is occupied
- Track temperature & lighting condition

* Uses previous information to control device

Adv: Handle partially observable environment & uses past

Dis:

Require additional memory & more complex design

3. Goal Based agent

A goal based agent select to achieve specific objective

* A goal based agent a room temperature
* Reduces electricity consumption
* Ensure home security

Adv: Support planning & flexible decision

Dis: Planning may require more computational resources

...artial information 3607
...l environment 3007

Disadvantage:

Cannot remember past event or lean user preferences

2. Model Based agent

A model based agent maintains as internal state using current is previous observation

- Remembers whether a room is occupied
- Track temperature is lighting condition

* Uses previous information to control device

Adv: Handle partially observable environment is uses past

Dis:

Require additional memory is more complex design

3. Goal Based agent

A goal based agent select to achieve specific objective

* A goal based agent a room temperature
* Reduces electricity consumption
* Ensure home security

Adv: Support planning is flexible decision

Dis: Planning may require more computational resources

partial information 75%
al environment 300%

4. Utlity Based Agent

A Utlity Based Agent evaluate different action & select one provide the greatest overall benefit

Adv: Balance multiple requirement & improve efficiency

Dis: Designing suitable utility function can be difficult

Comparsion:

Agent	Main Feature	Example
Simple reflex	Current input	Motion based
Model based	Internal state	Remembering
Goal based	Achieving goals	Maintaining
Utility based	Maximizing benefit	Balancing comfort

Conclusion:

A hybrid intelligent agent provide bette comfort, security & energy effiency than usin a single agent type

2. Uniformed Search Strategies for a Robot in a Unknown Maze

Introduction:

In Uniformed search algorithm explore possible path without knowing the location of the goal in advance Uniform cost search (UCS) help a robot find exit unknown maze

1. Uniform Cost search (UCS):

UCS select the node with the lowest total path cost

Working:

1. Start from the initial position with cost zero
2. Select the position with the lower cost
3. Check the whether it is the exit
4. If not explore neighbouring position & calculate their cost
5. Repeat untill the exit is found

Adv:

* Find the least-cost path
* Work well than movement cost differ

2. Uniformed Search Strategies for a Robot in a Unknown Maze

Introduction:

Uniformed search algorithm explore possible path without knowing the location of the goal in advance Uniform cost search

Dis : Require more memory

* May explore many unnecessary path

IDS (Iterative Deepening Search)

IDS perform depth limited search repeatedly, increase the depth

Working

* Start the a depth limit of zero
* Search all position within the limit
* If the exit is not found increase the limit
* Repeat untill the exit is reached

Complexity:

* UDS : Exponential time & pace
: IPS : Time $O(b^d)$, space $O(bd)$

Intelligent Agent:

A simple reflex agent react the obstacles a model based agent remember explored path. goal based agent plans to reach the exit, based agent cost and safety.

2 Conclusion:

UCS find the least cost path while IDS save memory.